

LegalMetriX — Platform Specification & Statutory Manual

End-to-End Legal Metrology Packaged Commodity Compliance, Inspection, Complaint & Enquiry Tracking System

Live Deployment URL: https://legal-metrology-dist-three.vercel.app/	Statutory Reference: Legal Metrology Act, 2009
Frontend Framework: React 18 + Vite + TypeScript + Tailwind v4	Compliance Standard: Packaged Commodities Rules, 2011
AI Assistance Engine: PaddleOCR + Gemini Vision (Dual-Pass)	Release Version: v2.4.0 (Enterprise Enforcement Suite)

1. Executive Summary & Core Mission

LegalMetriX is a dedicated statutory compliance, inspection auditing, complaint management, and enquiry tracking platform engineered to enforce the **Legal Metrology (Packaged Commodities) Rules, 2011** and the **Legal Metrology Act, 2009** across Indian retail and commercial markets.

The platform bridges physical ground seizures and digital statutory governance by integrating multi-modal optical package capture with automated declaration extraction, intelligent non-compliance flagging, departmental case escalation, authorized officer verification, and sanitized citizen grievance tracking.

2. System Architecture & Functional Modules

- **Multi-Modal Package Scanner:** Supports 3 flexible capture workflows: Single Image quick scan, 4-Panel Multi-Surface Capture (Front, Back, Left, Right with cropping and quality diagnostics), and 360° Rotational Continuous Video Capture with automated motion-blur rejection and sharp keyframe synthesis.
- **AI Evidence & Extraction Layer:** Employs an intelligent dual-pass PaddleOCR and Gemini Vision pipeline with strict anti-hallucination guardrails: unverified or smudged text is never guessed. AI results are strictly labeled PASS, POTENTIAL VIOLATION, or NEEDS VERIFICATION.
- **Statutory Rule Engine:** Evaluates all mandatory declarations under Rule 6(1) and Rule 12: Maximum Retail Price (MRP inclusive of taxes), Net Quantity with standard units, Complete Manufacturer/Packer Name & Address with PIN code, Manufacturing/Packing Date, Expiry/Best Before, Consumer Care Telephone & Email, and Country of Origin.
- **8-Stage Complaint & Enquiry Docket:** Manages formal legal enforcement proceedings from initial seizure to final closure across 8 standardized lifecycle stages: Submitted, Under Review, Further Enquiry, Awaiting Verification, Verified Violation, Not Verified, Action Taken, and Closed.
- **Multi-Persona Role Switcher:** Provides authenticated persona toggling between Inspector (field seizure and preliminary finding referral), Senior Official (statutory determination, legal notice issuance, laboratory transfer), and Citizen Consumer (public grievance tracking).
- **Citizen Public Tracking Portal:** Allows public consumers to track grievance dossiers by Complaint ID (e.g. LM-2026-XXXXXX). Implements mandatory privacy redaction, keeping internal enforcement notes, investigation memos, and officer contact details strictly confidential.
- **Compliance Assessment Reports:** Generates certified statutory assessment documents with declaration audit breakdown, rule non-conformance tags, evidence image bounding boxes, and official officer seal hashes (formally titled Inspection / Compliance Assessment Reports).

3. Statutory 8-Stage Complaint & Enquiry Lifecycle

Every potential violation detected during market sampling can be escalated into a formal Complaint Docket. The case progresses through an immutable, auditable statutory workflow:

Stage #	Status Name	Statutory Role & Enforcement Action	Public Visibility
1	Submitted	New complaint docket registered and linked to original Inspection ID.	Visible
2	Under Review	AI detection layer flags potential Rule 6 / Rule 12 non-conformances.	Visible
3	Further Enquiry	Escalated to Zonal Enforcement, Prosecution, or Laboratory Wings.	Visible (Safe Msg)
4	Awaiting Verification	Assigned to Senior Officer for statutory physical & legal determination.	Visible (Safe Msg)
5	Verified Violation	Officer confirms breach under Legal Metrology Act, 2009 (Sections 18/36).	Visible (Notice Issued)
6	Not Verified	Sample verified compliant upon physical caliper / measurement inspection.	Visible (Resolved)
7	Action Taken	Compounding fee recovered, seizure executed, or notice served.	Visible (Action Logged)
8	Closed	Case dossier officially archived with digital verification certificate.	Visible (Closed)

4. Statutory Decision-Making & AI Guardrails

Core Statutory Guardrail: Under Indian Administrative Law and the Legal Metrology Act, 2009, an Artificial Intelligence system cannot issue binding penal determinations. LegalMetriX enforces this principle at the architectural level:

- **AI Decision Separation:** AI model outputs are restricted to 'PASS', 'POTENTIAL VIOLATION', and 'NEEDS VERIFICATION'. They serve strictly as assistive cues for inspecting officers.
- **Authorized Human Sign-Off:** Only an authorized official (Inspector or Deputy Controller) holding valid statutory credentials can mark a case as 'VERIFIED VIOLATION', 'NOT VERIFIED', 'ACTION TAKEN', or 'CLOSED'.
- **Tamper-Evident Digital Seal:** Official determinations generate an immutable verification hash (e.g. SEAL-LM-DIR-XXXX) logged in the permanent audit trail.
- **No Misleading Legal Scores:** Overall compliance percentages are presented strictly as supporting diagnostic metrics rather than final legal judgments.

5. Technical Architecture & Endpoints

Layer	Technology	Description & Implementation
Frontend UI	React 18, TypeScript, Vite	Ultra-fast Single Page App with sub-400ms production builds.
Styling Engine	Tailwind CSS v4	High-contrast governmental theme (Navy #0f172a, Saffron #f59e0b).
Local Persistence	IndexedDB & LocalStorage	Resilient offline caching and automatic schema synchronization.
Offline PWA	Vite PWA / Workbox	Zero-downtime offline inspection recording with background sync.
API Services	FastAPI REST Ready	Structured endpoint client (/api/scans, /api/complaints, /api/reports).
Internationalization	Custom i18n Context	Full trilingual localization support in English, Hindi, and Telugu.

6. Codebase Architecture & File Structure

src/pages/ (Core Page Views)

- **HomePage.tsx:** Executive enforcement dashboard, 8-status metrics counters, 360° scanner spotlight.
- **ScanPage.tsx:** 4-stage inspection wizard (Upload, Keyframe Extraction, Review, Rule Verification & Escalation).
- **ComplaintsPage.tsx:** Complete complaint registry, status filtering, search bar, and clean empty state.
- **ComplaintDetailPage.tsx:** Comprehensive dossier with AI evidence ROI boxes, official verification, and audit logs.
- **TrackComplaintPage.tsx:** Public citizen grievance tracker with sanitized progress updates.
- **ReportsPage.tsx:** Official statutory inspection and assessment reports catalogue.
- **HistoryPage.tsx:** Field inspection history log with batch delete and report download options.

src/components/ (Interactive Statutory UI Elements)

- **RoleSwitcher.tsx:** Persona toggle (Inspector, Senior Official, Citizen).
- **ForwardModal.tsx:** Departmental case escalation dialog.
- **VerificationModal.tsx:** Statutory determination and digital seal signing dialog.
- **EvidenceModal.tsx:** Cropped photographic evidence and OCR ROI bounding box inspector.
- **NewComplaintModal.tsx:** Manual complaint docket filing modal.
- **Navigation.tsx:** Top navigation bar with language picker, role badge, and route links.

src/services/ & src/types/ (Business Logic & Types)

- **complaintService.ts:** Data store, CRUD, timeline generator, and public tracking sanitizer.
- **api.ts:** REST API client abstraction.
- **complaint.ts:** Core TypeScript data models.

7. Operational Instructions for Demonstration

- **Conducting a Fresh Inspection:** Navigate to '/scan' -> Upload packaging images or record 360° video -> The AI pipeline extracts mandatory declarations -> If any Rule 6 breach occurs, click 'Create Complaint / Enquiry' to generate an official LM-2026-XXXXXX dossier.
- **Escalating to Higher Authority:** Open the Complaint Dossier -> Click 'Forward for Further Enquiry' -> Select target authority (Prosecution Cell, Laboratory Wing) and specify grounds -> Case transitions to 'Further Enquiry'.
- **Official Statutory Verification:** Switch persona to 'Senior Official' -> Open Complaint Dossier -> Click 'Official Verification' -> Select verdict ('Verified Violation' / 'Action Taken') -> Enter statutory observations -> Seal with digital signature hash.
- **Citizen Public Verification:** Navigate to '/track' -> Search using the Complaint ID -> Review sanitized, tamper-evident progression timeline without exposing confidential notes.